



UNIVERSITY OF MADRAS

Established under the Act of Incorporation XXVII of 1857- Madras University Act 1923]

[State University]

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Heavy mineral and Petrography Report

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Abstract

This report investigates the heavy mineral distribution in two sedimentary samples collected from distinct locations in Ghana. The study aims to analyze the mineralogical composition, identify provenance indicators, and interpret depositional environments. Petrographic examination and heavy liquid separation techniques were employed, followed by microscopic identification and point counting of the mineral assemblages. Results indicate distinct mineralogical signatures with implications for sediment source and geological history.

1. Introduction

Heavy minerals are denser than 2.89 g/cm^3 and are found in minor amounts in most sedimentary rocks. Despite their low abundance, they serve as valuable indicators of provenance, tectonic setting, and sedimentary processes. Ghana, located within the West African Craton, presents an ideal setting for heavy mineral studies due to its diverse geological units, including Birimian rocks and the Tarkwaian system.

This report compares two sedimentary samples from Ghana:

Sample I : Hubert

Sample II: Salai

The objective is to characterize and compare the heavy mineral assemblages of these two samples to understand their provenance and depositional history.

2. Materials and Methods

2.1 Sample Collection

Samples were collected from surficial sediments at a depth of 0–50 cm. Sample (I and II) was retrieved from a fluvial environment.

2.2 Sample Preparation

Drying and Sieving: Samples were air-dried and sieved to isolate the 60, 120, 230, 230,µm grain-size fraction.

Heavy Mineral Separation: The sieved fraction was subjected to heavy liquid separation using bromoform (specific gravity 2.89).

Mounting and Microscopy: Recovered heavy mineral concentrates were mounted on glass slides using Canada balsam and examined under a polarizing microscope.

2.3 Mineral Identification and Point Counting

A minimum of 300 grains per slide were counted using the ribbon method. Minerals were identified based on optical properties, and data were tabulated to determine modal compositions.

3. Results

3.1 Sample I

Dominant Minerals are:

1. Magnetite
2. Ilmenite
3. Zircon
4. Rutile
5. Garnet
6. Gold flakes

Grain Morphology: Well-rounded to sub-rounded, suggesting prolonged transport and reworking.

3.2 Sample II

Dominant Minerals are:

1. Magnetite
2. Ilmenite
3. Zircon
4. Rutile
5. Garnet
6. Gold flakes

Grain Morphology: Sub-angular to angular, indicating limited transport and possible local source.

4. Discussion

Study Area Description

1.Sample.I&II

Ghana's major rivers, draining the forested central and western regions. The sampling location falls within a fluvial environment where sediment transport is largely controlled by seasonal rainfall and river discharge. Birimian Supergroup, known for its ancient volcanic and plutonic rocks, which are potential sources of stable heavy minerals such as zircon, tourmaline, and rutile. The area has experienced extensive weathering and sediment recycling, contributing to a mature sedimentary load with well-rounded mineral grains. This location sits near the transition zone between the Dahomeyide Belt and the Volta Basin, where varied lithologies contribute to a diverse sediment supply. Short-distance fluvial and coastal transport processes lead to the deposition of angular to sub-angular heavy minerals. The influence of coastal currents, wave reworking, and riverine input from the Volta River makes this site an important intersection of terrestrial and marine sediment sources.

Heavy Mineral Descriptions

Magnetite (Fe_3O_4)

Magnetite is a strongly magnetic opaque mineral, often found in mafic and ultramafic rocks. Its significant presence in Sample I and II suggests derivation from nearby igneous terrains. It is easily distinguished under a microscope due to its opaque black color and strong magnetism.

Ilmenite ($FeTiO_3$)

Ilmenite is an opaque, metallic black mineral commonly found in both igneous and metamorphic rocks. It is dominant in Sample I and II and occurs as sub-angular grains. Its high density and magnetic properties allow easy recovery during separation. It is a key indicator of mafic source rocks.

Zircon ($ZrSiO_4$)

Zircon is a mineral belonging to the group of nesosilicates and is a source of the metal zirconium. Its chemical name is zirconium, Zircon precipitates from silicate melts and has relatively high concentrations of high field strength incompatible elements. For example, hafnium is almost always present in quantities. The crystal structure of zircon is tetragonal crystal system. The natural color of zircon varies between colorless, yellow-golden, red, brown, blue, and green.

Rutile (TiO_2)

Rutile is an oxide mineral composed of titanium dioxide (TiO_2), the most common natural form of TiO_2 . Rarer polymorphs of TiO_2 are known, including anatase, akaogiite, and brookite. Rutile has one of the highest refractive indices at visible wavelengths of any known crystal and also exhibits a particularly large birefringence and high dispersion. Natural rutile may contain up to 10% iron and significant amounts of niobium and tantalum.

Garnet (Fe, Mg, Mn, Ca silicate)

Garnet appears in both samples. It typically forms sub-rounded to angular grains and varies in color from pink to reddish-brown. Garnet is derived from medium- to high-grade metamorphic rocks, and its angular nature in coastal sediments indicates short transport distances.

Gold (Au)

Gold is a chemical element; it has chemical symbol Au (from Latin aurum) and atomic number 79. In its pure form, it is a bright, slightly orange-yellow, dense, soft, malleable, and ductile metal. Chemically, gold is a transition metal, a group 11 element, and one of the noble metals. It is one of the least reactive chemical elements, being the second-lowest in the reactivity series. It is solid under standard conditions. Gold often occurs in free elemental (native state), as nuggets or grains, in rocks, veins, and alluvial deposits. It occurs in a solid solution series with the native element silver (as in electrum), naturally alloyed with other metals like copper and palladium, and mineral inclusions such as within pyrite.

Procedure

The following procedure have been adopt

2.2 Sample Preparation

Drying and Sieving: Samples were air-dried and sieved to isolate the 63–125 μm grain-size fraction.

Heavy Mineral Separation: The sieved fraction was subjected to heavy liquid separation using bromoform (specific gravity 2.89).

Mounting and Microscopy: Recovered heavy mineral concentrates were mounted on glass slides using Canada balsam and examined under a polarizing microscope.

Sieve analysis

About 150 -200 grams of each sample were taken in porcelain dishes and air-dried, taking all possible care against contamination. Sandy samples were repeatedly washed in fresh water for the removal of salts and then dried. After drying, a sub-sample weighing about 35 – 50 grams was obtained by coning and quartering. The sub-sample was subjected to sieving by ASTM test sieves of 8’’ diameter with successive sieves spaced at $\frac{1}{2}$ phi intervals. Material about 1mm in size was hand-sieved, and those less than 1mm in size were subjected to sieving on a Ro-tap sieve shaker for 15 minutes. Material thus retained on each sieve, was weighed on Mettler balance.

Methodology for Heavy Mineral Studies

A total of 54 samples representing the entire area covering all sedimentary facies were selected and processed for heavy mineral studies. The procedure outlined by Carver (1971) was followed for heavy mineral studies. The samples were wet-sieved through +230 ASTM mesh. The +230 mesh (0.062 mm) fraction was treated with 10 % dilute HCl to remove acid-digestible material. Then it was divided by dry sieving into +60 (+0.25 mm), -60 to +120 (- 0.25 to +0.125 mm) and -120 to +230 (-0.125 to +0.062 mm) ASTM mesh fractions. Heavy minerals were separated from those fractions using bromoform (sp.gr.2.89). The weight percentages of heavy and light minerals in each fraction and in the total sample were calculated and tabulated. As the weight percentage of heavies is negligible in +0.25mm fraction of the sediment of the area, the other two size fractions viz. -0.25 to +0.125mm and -0.125 to +0.062 mm were chosen for detailed quantitative study as the heavy minerals are associated with medium to fine-grained size sediments. Identification and counting of the grains were carried out under a Lica petrological microscope fitted with a mechanical stage. About 300 to 400 grains were counted on each slide following the line method (Galehouse, 1971) and the number percentage of heavy minerals in - 0.25 mm to +0.125 mm and -0.125 mm to +0.062 mm size fractions were calculated and tabulated. Areal distribution maps of heavy minerals in each fraction were prepared using the number percentage of minerals in the two factions. Based on the distribution pattern, major mineral assemblages were identified. An attempt has been made to decipher the direction of mineral dispersal based on the distribution of medium and fine-size fractions of heavy minerals.

CONCLUSION

The following minerals were identified

1. Magnetite – 1.722%
2. Ilmenite – 0.162%
3. Zircon – 0.01%
4. Rutile – 0.005%
5. Gold flakes – 3.20PPM